

IEEE 830 Software Requirements Specification (SRS)

Project: SIM Web App Solution

1. Introduction

Purpose: Define the functional and non-functional requirements for a web-based Student Information Management (SIM) application.

Scope: The system enables administrators to manage student records and students to securely access their information.

Definitions: SIM = Student Information Management.

2. Overall Description

Product Functions: Register/Login, Dashboard, Manage Student Records, Search Records, Generate Reports, Logout.

User Classes: Administrator, Student.

Operating Environment: Modern web browser, internet connection, server with database.

Constraints: Secure authentication, responsive UI, role-based access.

3. Specific Requirements

Functional Requirements: User authentication; CRUD student records; search/filter; report generation; profile management; audit logging.

Non-Functional Requirements: Response time under 3 seconds for common operations; encrypted passwords; 99% availability target; scalable architecture; usability and accessibility.

External Interfaces: Browser UI, relational database.

Database Requirements: Tables for Users, Students, Courses, Results, Audit Logs.

Security: Authentication, authorization, hashed passwords, input validation, session timeout.

Quality Attributes: Reliability, maintainability, portability, performance.